# Seok Jung Kim

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## Research Interests

- Drone vehicle routing and last-mile delivery under urban wind
- Hybrid metaheuristics (VNS, ALNS) for large-scale routing problems
- Learning-assisted optimization (reinforcement learning within metaheuristics)

## Education

**Kyung Hee University** Sep. 2026 – Present
*M.S. in Industrial and Management Systems Engineering (IO-LAB)* Yongin, Korea

**Chonnam National University** Mar. 2023 – Aug. 2025
*B.B.A. in Business Administration / Double Major in Big Data Convergence* Gwangju, Korea

## Research Experience

**Wind- and Level-aware Drone Vehicle Routing Problem (WiLD-VRP)** Sep. 2026 – Present
*Advisor: Prof. Ho Young Jeong*

- Formulate a multi-trip drone routing problem in which the flight level of each route segment is chosen jointly with the visiting sequence under altitude-dependent urban wind.
- Develop a variable neighborhood search (VNS) with an offline Q-learning rule that skips unpromising move evaluations; benchmark against SA, ALNS, and exact MILP solutions (Gurobi).
- Build test instances from real building data for five districts of Seoul.
- Manuscript in preparation for *Transportation Research Part E*.

## Presentations

**S. J. Kim** and H. Y. Jeong, "Segment-wise altitude selection in urban drone delivery routing under wind," Asia Pacific Industrial Engineering & Management Systems Conference (APIEMS 2026), Busan, Republic of Korea, Dec. 2026 (scheduled, oral).

## Honors & Awards

**Encouragement Award**, The 4th Joint Academic Seminar, Kyung Hee University Nov. 2026
*For a presentation of the WiLD-VRP research*

## Activities and Experience

**KDI Knowledge Sharing Program (KSP)** 2024 – 2025
*Young KSP (YKSP) Cohort 8 – Infrastructure & Transportation Information Systems, Honduras; two field dispatches (2024, 2025)*

**Korea–Arab Youth Exchange Program** 2025
*Sponsored by S-OIL; field exchange to the UAE (Dubai) and Saudi Arabia*

**World Friends Korea (WFK) Online Education Volunteer** 2025
*Online education for students in Madagascar*

**LG Aimers Cohort 6** 2024
*AI/ML Education Program, LG AI Research*

## Certifications & Skills

- **Programming & Tools**: Python, C++, Gurobi
- **Methods**: Mixed-integer linear programming (MILP), variable neighborhood search (VNS), adaptive large neighborhood search (ALNS), reinforcement learning (Q-learning)
- **Certification**: ADsP (Advanced Data Analytics Semi-Professional), 2024
- **Language**: TOEIC 910 (ETS), 2024